

Problem Statement:

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all digits are even.
 - E.g. 1: Number: 12345, Output: All digits not even
 - E.g. 2: Number: 24680, Output: All digits are even
- Check whether all digits are prime.
 - E.g. 1: Number: 2357, Output: All digits are prime
 - E.g. 2: Number: 2379, Output: All digits are not prime

ALGORITHM**Main Algorithm:**

1. Start
2. Display menu:
 1. Check whether all digits are even
 2. Check whether all digits are prime
 3. Exit
3. Read choice from user.
4. If `choice == 1` or `choice == 2`:
 Prompt user and read positive integer `num`.
5. Using `switch(choice)`:
 - **Case 1:** Call `areAllDigitsEven(num)`. If true, print "All digits are even", else print "All digits not even".
 - **Case 2:** Call `areAllDigitsPrime(num)`. If true, print "All digits are prime", else print "All digits are not prime".
 - **Case 3:** Exit program.
 - **Default:** Print "Invalid choice!".
6. Stop

Function: `areAllDigitsEven(num)`

1. If `num == 0`, return 1.
2. While `num > 0`:
 - `rem = num % 10`
 - If `rem % 2 != 0`, return 0.
 - `num = num / 10`

3. Return 1.

Function: areAllDigitsPrime(num)

1. While num > 0:

- rem = num % 10
- If rem != 2 AND rem != 3 AND rem != 5 AND rem != 7, return 0.
- num = num / 10

2. Return 1.

TEST CASE TABLE

Test Case ID	Input Number	Choice	Expected Output	Actual Output	Result
TC_01	12345	1	All digits not even	All digits not even	Pass
TC_02	24680	1	All digits are even	All digits are even	Pass
TC_03	2357	2	All digits are prime	All digits are prime	Pass
TC_04	2379	2	All digits are not prime	All digits are not prime	Pass

C PROGRAM

```
#include <stdio.h>

// Function to check if all digits in a number are even
int areAllDigitsEven(int num) {
    int rem;
    if (num == 0) return 1;

    while (num > 0) {
        rem = num % 10;
        if (rem % 2 != 0) {
            return 0; // Found odd digit
        }
        num /= 10;
    }
    return 1;
}

// Function to check if all digits in a number are prime
int areAllDigitsPrime(int num) {
    int rem;
    while (num > 0) {
        rem = num % 10;
        if (rem != 2 && rem != 3 && rem != 5 && rem != 7) {
            return 0; // Found non-prime digit
        }
        num /= 10;
    }
    return 1;
}
```

```

int main() {
    int choice, num;

    printf("
--- MENU ---
");
    printf("1. Check if all digits are even
");
    printf("2. Check if all digits are prime
");
    printf("3. Exit
");
    printf("Enter choice: ");
    scanf("%d", &choice);

    if (choice == 1 || choice == 2) {
        printf("Enter a positive integer: ");
        scanf("%d", &num);
    }

    switch (choice) {
        case 1:
            if (areAllDigitsEven(num))
                printf("All digits are even
");
            else
                printf("All digits not even
");
            break;

        case 2:
            if (areAllDigitsPrime(num))
                printf("All digits are prime
");
            else
                printf("All digits are not prime
");
            break;

        case 3:
            printf("Exiting...
");
            break;

        default:
            printf("Invalid choice!
");
    }

    return 0;
}

```

PROGRAM EXECUTION (TERMINAL OUTPUT)

```

gcc solution.c -o solution
./solution

```

```
--- MENU ---
1. Check if all digits are even
2. Check if all digits are prime
3. Exit
Enter choice: 1
Enter a positive integer: 12345
All digits not even
```

./solution

```
--- MENU ---
1. Check if all digits are even
2. Check if all digits are prime
3. Exit
Enter choice: 1
Enter a positive integer: 24680
All digits are even
```

./solution

```
--- MENU ---
1. Check if all digits are even
2. Check if all digits are prime
3. Exit
Enter choice: 2
Enter a positive integer: 2357
All digits are prime
```

./solution

```
--- MENU ---
1. Check if all digits are even
2. Check if all digits are prime
3. Exit
Enter choice: 2
Enter a positive integer: 2379
All digits are not prime
```